INCH-POUND

MIL-M-28787/250
AMENDMENT 2
28 April 1993
SUPERSEDING
AMENDMENT 1
13 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, DIGITAL, KEY CODE AMF

This amendment forms a part of Military Specification MIL-M-28787/250(EC), dated 17 March 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, 6, AND 7

TABLE I, Class II column heading: Delete "100°" (four places) and substitute "85°" (four places).

PAGE 8

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through JJ shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, and C where applicable, followed by the remaining tests in the sequence."

*4.5.1, fifth line: Delete "B through K," and substitute "D through K,".

*4.5.3, third line: Delete "B through K," and substitute "D through K,".

*4.5.3, sixth line: Delete "B through JJ" and substitute "D through JJ".

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PAGE 9

6.4, second line: Delete "100°C" and substitute "85°C".

6.4, Output drive current (V_{DD} = 5V): Delete "100°C" (two places) and substitute "85°C" (two places).

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FIGURE 6: Delete and substitute new figure 6 as printed on page 4 of this amendment.

PAGE 19

*TABLE VI, NOTE 3, first line: Delete "voltmeter V03" and substitute "item M02".

PAGE 20

FIGURE 7: Delete and substitute new figure 7 as printed on page 5 of this amendment.

PAGE 23

***FIGURE 9:** Delete and substitute new figure 9 as printed on page 6 of this amendment.

PAGE 25

FIGURE 10: Delete and substitute new figure 10 as printed on page 7 of this amendment.

PAGE 28

FIGURE 12: Delete and substitute new figure 12 as printed on page 8 of this amendment.

PAGE 30

FIGURE 13: Delete and substitute new figure 13 as printed on page 9 of this amendment.

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FIGURE 15: Delete and substitute new figure 15 as printed on page 10 of this amendment.

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FIGURE 19: Delete and substitute new figure 19 as printed on page 11 of this amendment.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

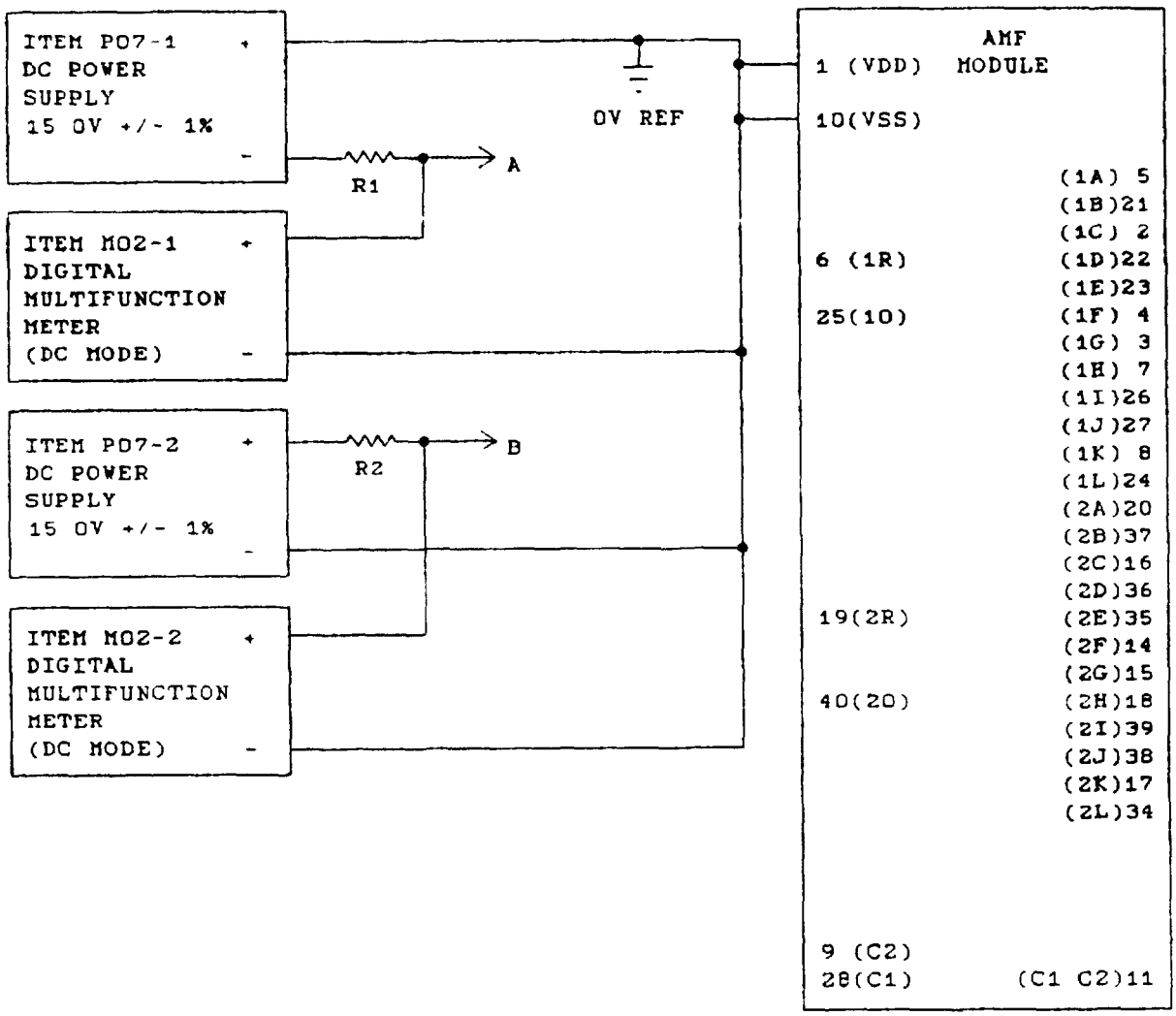
Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0202)

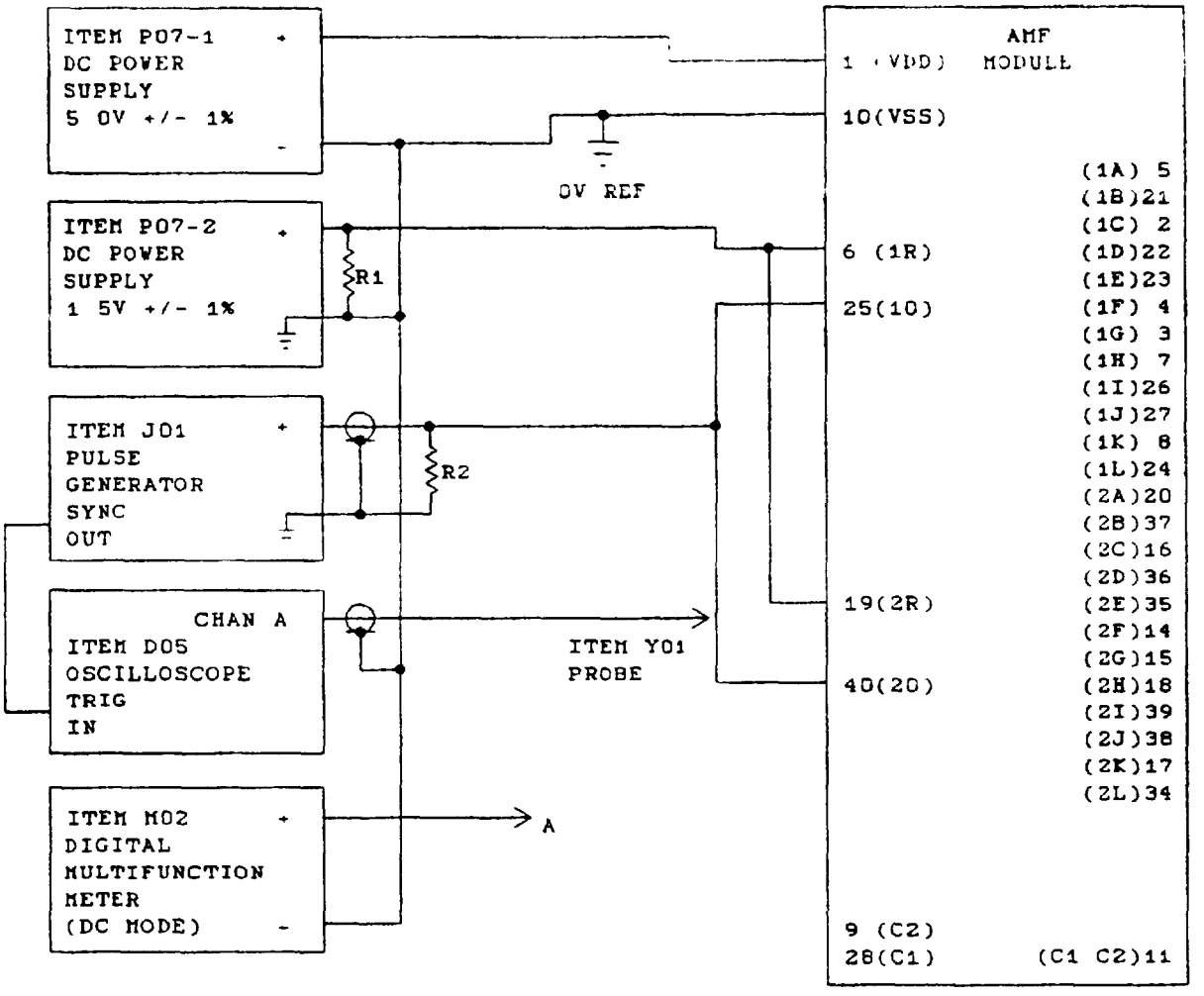
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NOTES:
1. R1 = 9.09 kΩ ± 1%, 1/8 W.
2. R2 = 13.7 kΩ ± 1%, 1/8 W.
3. Since items P07-1, P07-2, M02-1, and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

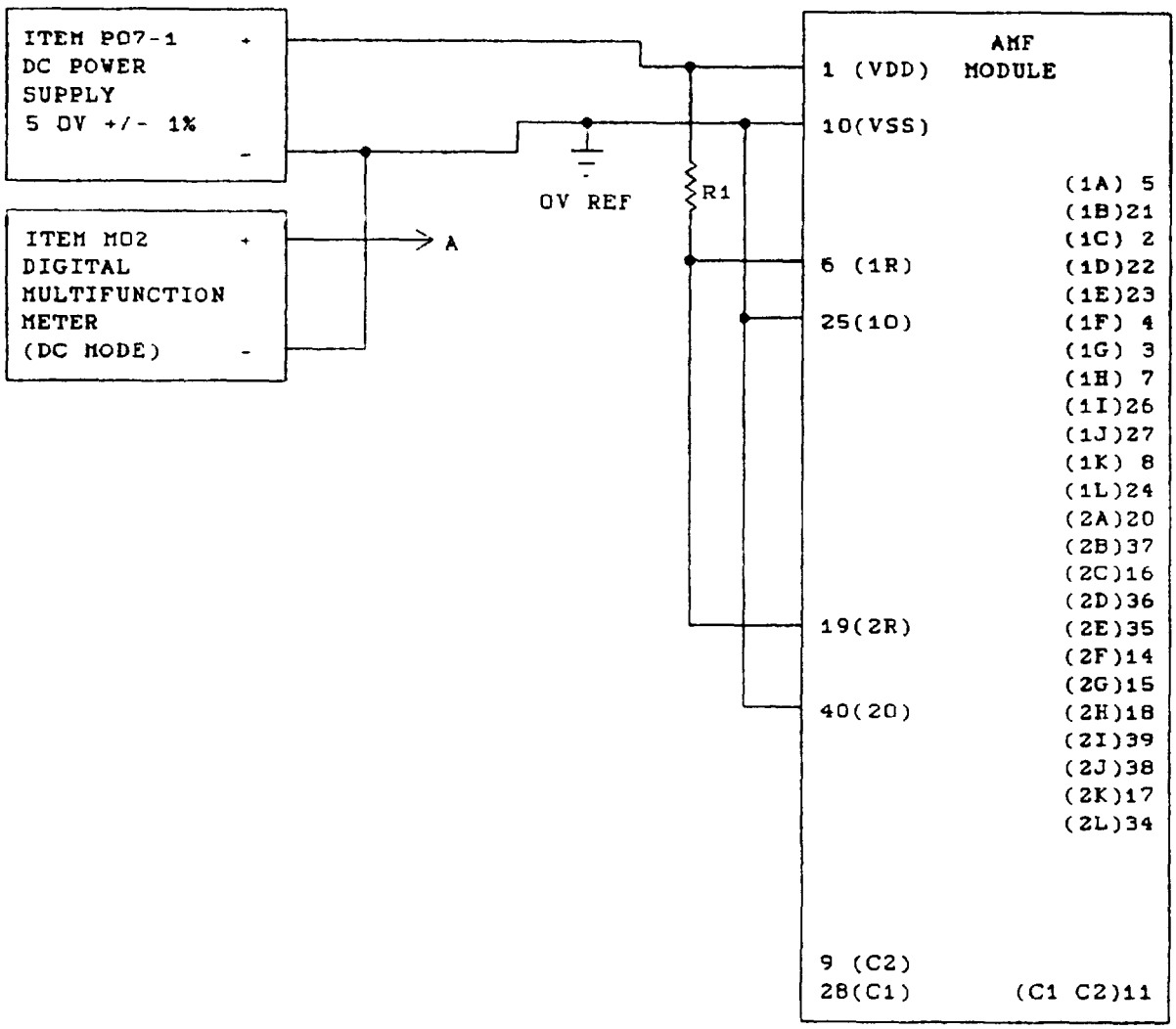
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NOTE: R1 = R2 = 51.1 Ω ± 5%, 1/2 W.

FIGURE 7. Functional test circuit for table VI.

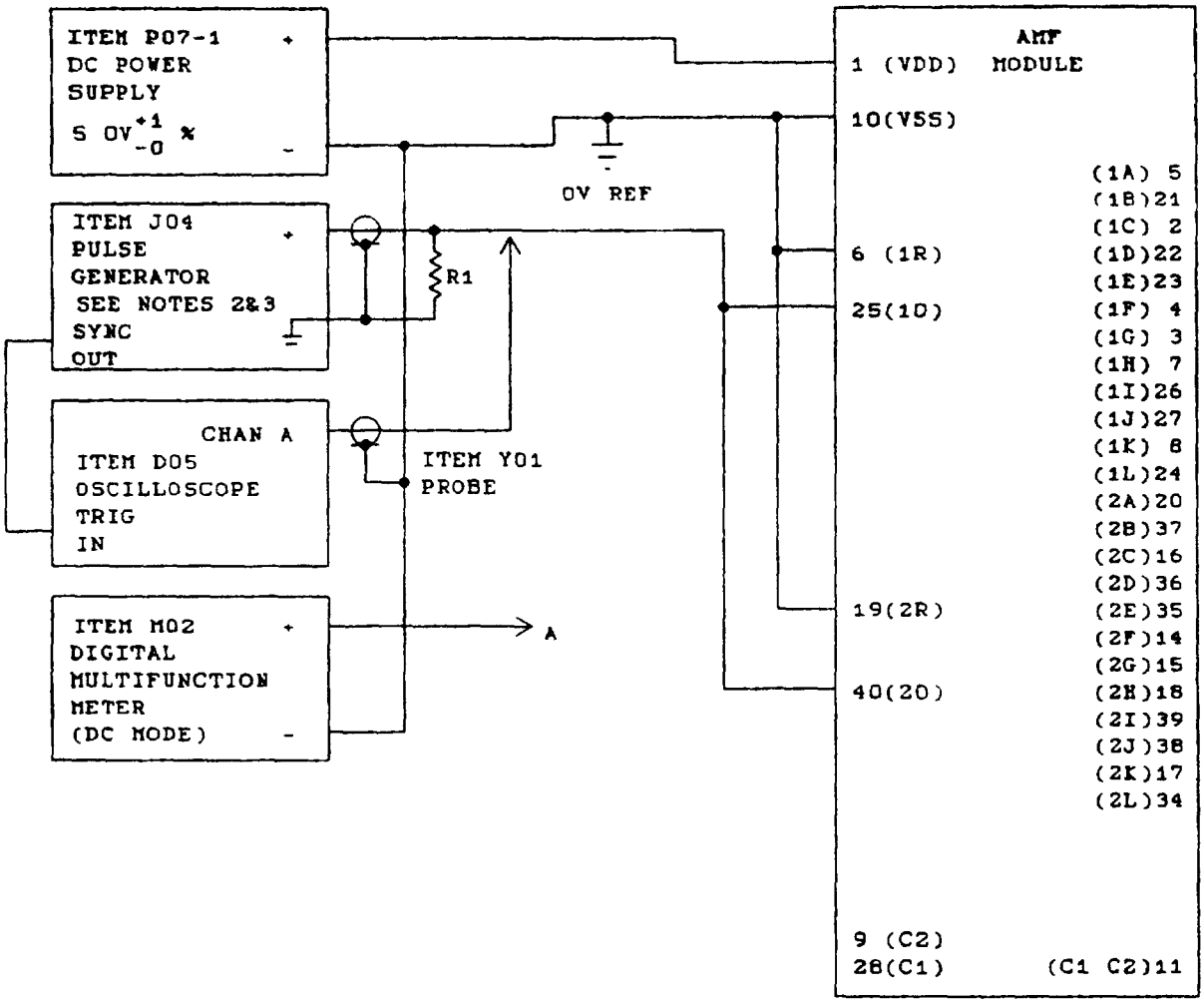
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NOTE: R1 = 1.5 kΩ ± 1%, 1/4 W.

FIGURE 9. Static output voltage (low-level) test circuit for table VII.

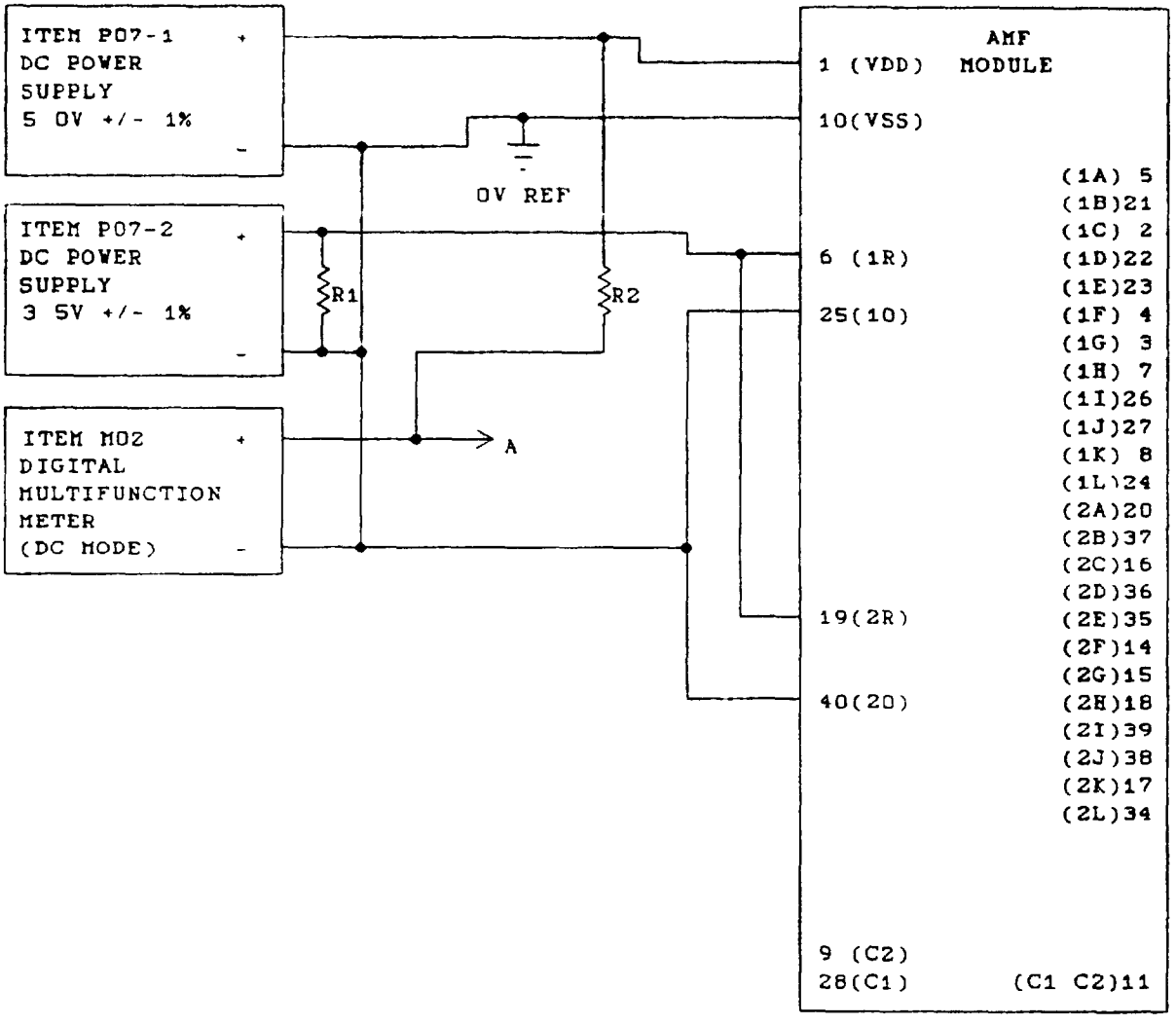
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NOTES:
1. **R1 = 51.1 Ω ± 5%, 5 W.**
2. **Before pulsing J04 4095 times, momentarily reset pins 6 and 19 high.**
3. **After making all connections, turn item J04 on and allow only 4095 clock pulses to pass and then turn item J04 off.**

FIGURE 10. Static output voltage (high-level) test circuit for table VIII.

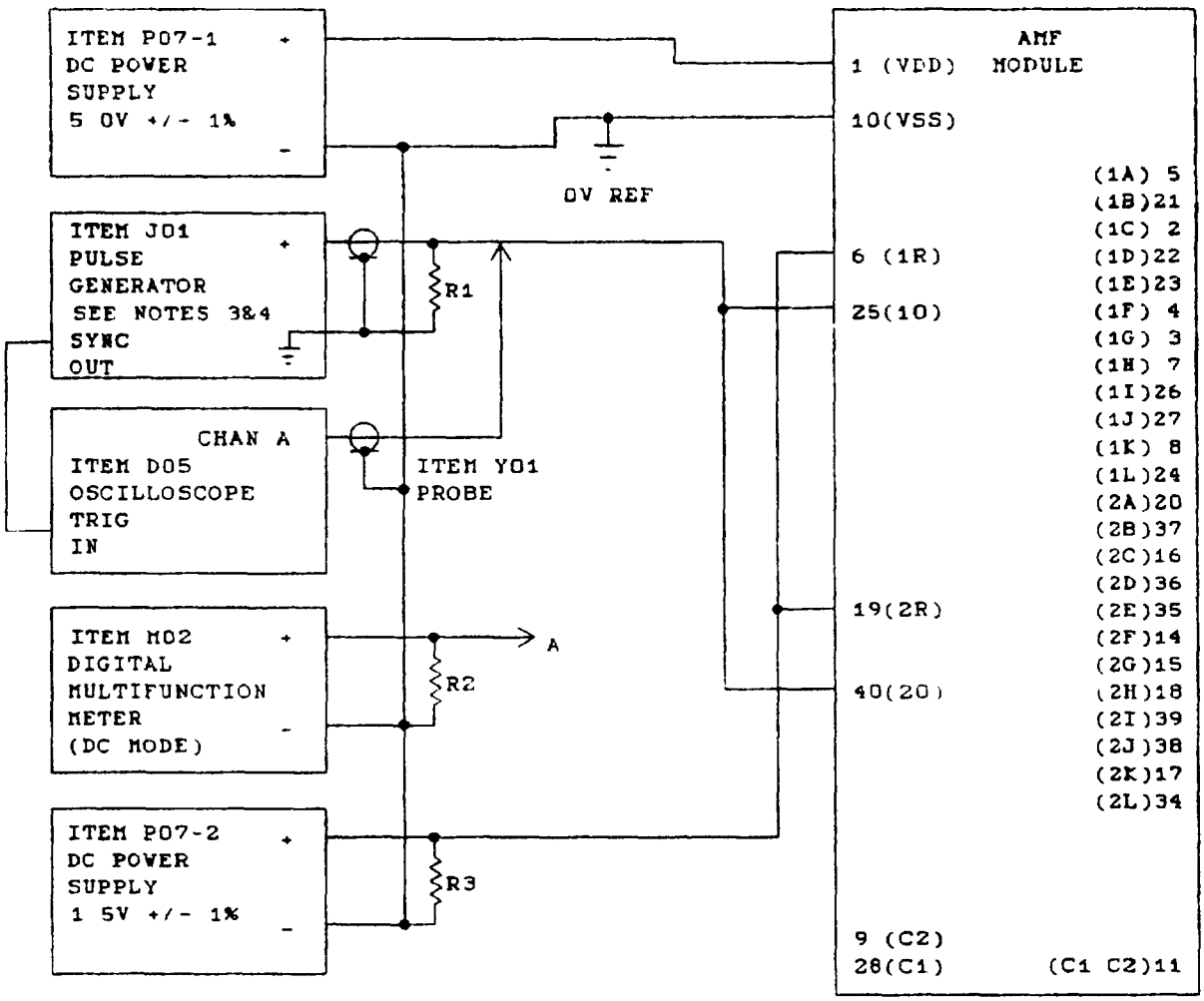
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NOTES:
1. R1 = 51.1 Ω ± 5%, 1/2 W.
2. R2 = @ -55°C 7.15 kΩ ± 1%, 1/8 W.
   = @ 0°C and 25°C 9.09 kΩ ± 1%, 1/8 W.
   = @ 60°C 10.5 kΩ ± 1%, 1/8 W.
   = @ 85°C 12.7 kΩ ± 1%, 1/8 W.

FIGURE 12. Static output voltage (low-level, VDD = 5.0 V) test circuit for table IX.

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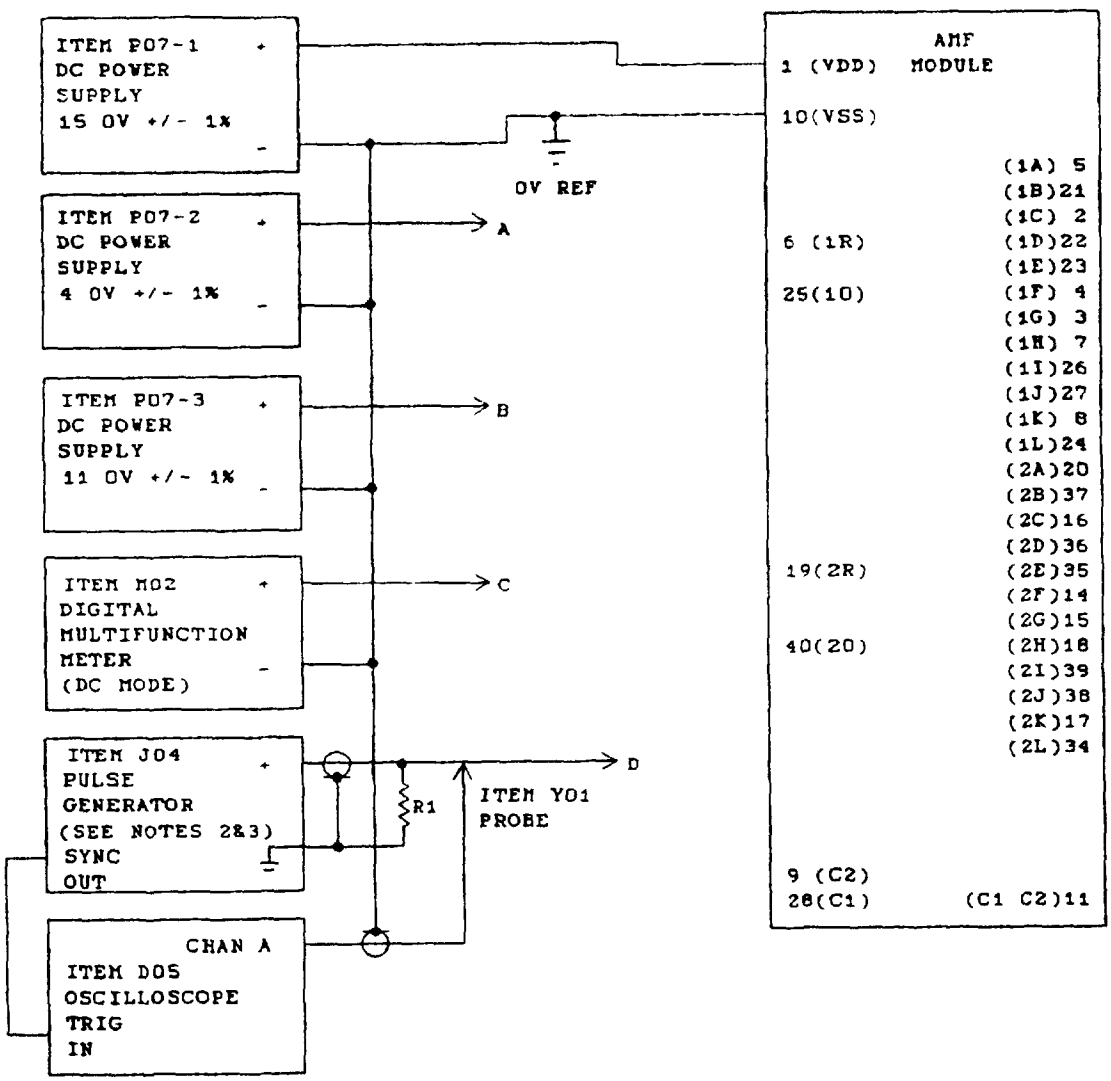


NOTES:

1. R1, R3 = 51.1 Ω ± 5%, 1/2 W.
2. R2 = @ -55°C 18.2 kΩ ± 1%, 1/8 W.
   = @ 0°C and 25°C 23.2 kΩ ± 1%, 1/8 W.
   = @ 60°C 27.4 kΩ ± 1%, 1/8 W.
   = @ 85°C 32.4 kΩ ± 1%, 1/8 W.
3. Before pulsing J01 4095 times, momentarily reset pins 6 and 19 high.
4. After making all connections, turn item J01 on and allow only 4095 clock pulses to pass and then turn item J01 off.

FIGURE 13. Static output voltage (high-level, VDD = 5 V) test circuit for table X.

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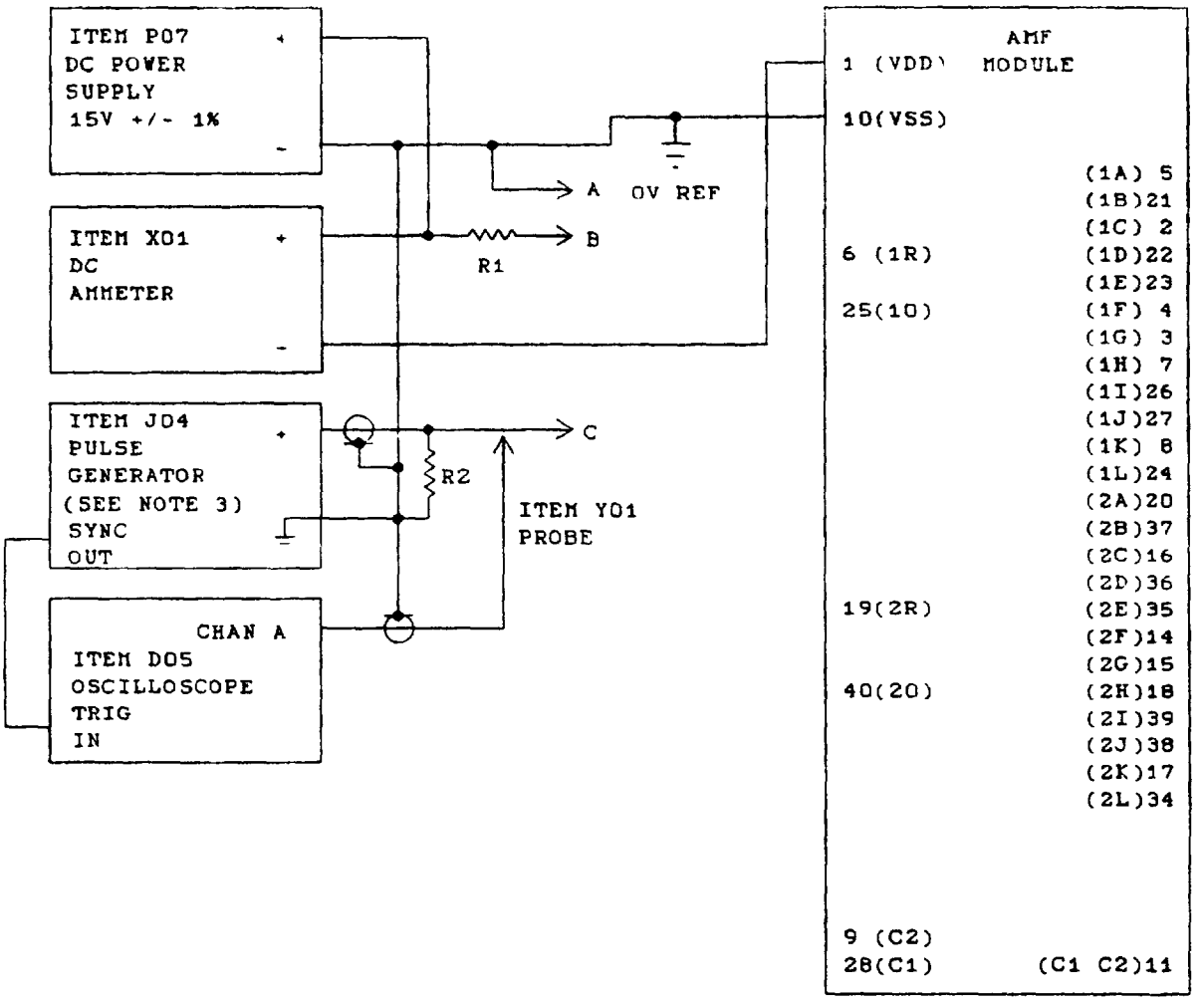


NOTES:

1. R1 = 51.1 Ω ± 5%, 5 W.
2. For test conditions K1-K24, momentarily reset pins 6 and 19 high before connecting point A and pulsing J04 4095 times.
3. For test conditions K1-K24, allow only 4095 clock pulses to pass and then turn item J04 off before making measurements.
4. All voltages are referenced to ground.

FIGURE 15. Static output voltage (low-level and high-level, VDD = 15 V) test circuit for tables XI and XII.

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CAUTION: Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 1.5 kΩ ± 1%, 1/4 W.
2. R2 = 51.1 Ω ± 5%, 5W.
3. Allow only 4095 clock pulses to pass after connecting point C, then turn item J04 off before measuring test condition N2.

FIGURE 19. Power supply current (low-level and high-level) test circuit for table XV.

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